

CORE JAVA COMPLETE NOTES

1. Introduction to Java

- Java is a high-level, object-oriented programming language.
- Platform independent using JVM.
- Features: Simple, Robust, Secure, Portable, Multithreaded, High Performance.

2. JVM, JRE, JDK

- JVM: Executes Java bytecode.
- JRE: JVM + Libraries (runs Java apps).
- JDK: JRE + Development tools (compilers etc.)

3. Data Types

- Primitive: byte, short, int, long, float, double, char, boolean.
- Non-primitive: String, Arrays, Classes.

4. Variables

- Instance Variable
- Static Variable
- Local Variable

5. Operators

- Arithmetic, Relational, Logical, Bitwise, Assignment, Unary, Ternary.

6. Control Statements

- if, if-else, nested-if, switch.
- Loops: for, while, do-while.
- Jump: break, continue, return.

7. OOP Concepts

- Class & Object
- Constructor (Default, Parameterized)

- Encapsulation
- Inheritance (Single, Multilevel, Hierarchical)
- Polymorphism (Compile-time & Runtime)
- Abstraction (Abstract class & Interface)

8. Packages

- Built-in & user-defined packages.
- import keyword.

9. Arrays

- 1D, 2D, Jagged arrays.
- Array methods.

10. Strings

- Immutable class.
- Important methods: length(), charAt(), substring(), trim(), toUpperCase() etc.
- StringBuilder & StringBuffer.

11. Exception Handling

- try, catch, finally, throw, throws.
- Checked vs Unchecked exceptions.
- Custom exception.

12. File Handling

- FileInputStream, FileOutputStream.
- FileReader, FileWriter.
- Buffered classes.

13. Collections Framework

- List: ArrayList, LinkedList.
- Set: HashSet, LinkedHashSet, TreeSet.

- Map: HashMap, LinkedHashMap, TreeMap.
- Iterator & ListIterator.

14. Multithreading

- Thread class & Runnable interface.
- Thread lifecycle.
- Synchronization.
- Inter-thread communication.

15. JDBC Basics

- Driver loading.
- Connection, Statement, ResultSet.
- CRUD operations.

16. Wrapper Classes

- Integer, Double, Boolean etc.
- Autoboxing & Unboxing.

17. Inner Classes

- Member, Static, Local, Anonymous.

18. Object Class Methods

- toString(), equals(), hashCode(), clone(), finalize().

19. Access Modifiers

- private, default, protected, public.

20. Java Memory Area

- Stack, Heap, Method Area, PC Register.

21. Garbage Collection

- finalize(), automatic memory management.

22. Keywords

- this, super, final, static, abstract, synchronized, volatile, transient.

23. Interfaces vs Abstract Classes

24. Enum

- User-defined constants.

25. Java 8 Features

- Lambda Expressions
- Functional Interfaces
- Stream API
- Default & Static methods in interface

THIS PDF COVERS ALL IMPORTANT CORE JAVA TOPICS FOR EXAM & INTERVIEW.